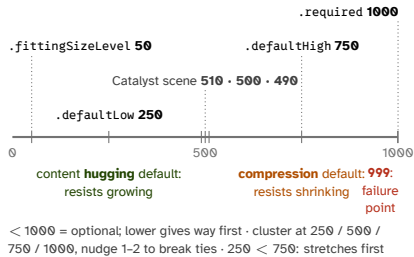


Named facts — the numbers, names and lists of iOS development

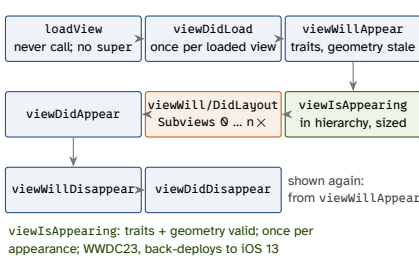
recall · folio

In one line: the exact values, case lists and call orders that UIKit, Foundation, SwiftUI and the test frameworks fix by definition, and the names that patterns and principles were given by their authors — each checked against Apple's documentation, the Swift Book or the originator. The why behind each lives in the linked folios.

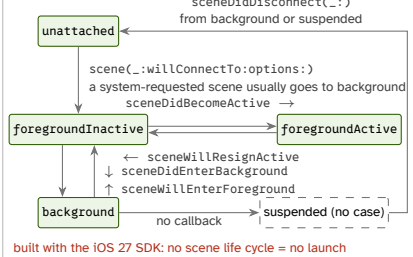
UILayoutPriority (Float, 1 ... 1000)



UIViewController — appearance



UIScene.ActivationState

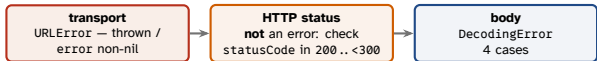


Auto Layout

name	value / rule
installed priority autoresizing mask	changing it to or from .required after the constraint is added throws translatesAutoresizingMaskIntoConstraints: true for a view made in code, false from Interface Builder, leave it alone on arrangedSubviews, cells, a VC's view
conflict	engine breaks constraints until the layout is valid, logs; breakpoint UIAlertForUnsatisfiableConstraints; never required hugging / compression
self-sizing rows	rowHeight and estimatedRowHeight default to .automaticDimension; an IB cell needs rowHeight = .automaticDimension in code; estimate ≠ 0; heightForRowAt overrides and runs for every row (Apple: hurts at 1000+)
size class	UIUserInterfaceSizeClass: .compact · .regular · .unspecified, per axis; set by device, window, multitasking

UIStackView	default	cases
axis	.horizontal	.horizontal · .vertical
distribution (along axis)	.fill	.fill · .fillEqually · .fillProportionally · .equalSpacing · .equalCentering
alignment (across)	.fill	.fill · .leading (= .top) · .firstBaseline · .center · .trailing (= .bottom) · .lastBaseline
spacing	0	a minimum for the two equal kinds; negative overlaps; spacingUseSystem (iOS 11)

Networking & Codable



URLSession reports only client-side errors; a 404 arrives as a response (as? HTTPURLResponse).

URLError.Code (NSErrorDomain)	
-999 cancelled	-1009 notConnectedToInternet
-1000 badURL	-1011 badServerResponse
-1001 timeout	-1020 dataNotAllowed (cellular refused)
-1003 cannotFindHost	-1022 appTransportSecurityRequires SecureConnection
-1004 cannotConnectToHost	-1200 secureConnectionFailed
-1005 networkConnectionLost	-1202 serverCertificateUntrusted

DecodingError: .typeMismatch(type, ctx) · .valueNotFound(type, ctx) (null for a non-optional) · .keyNotFound(key, ctx) · .dataCorrupted(ctx); Context = codingPath, debugDescription, underlyingError; EncodingError: only .invalidValue.
Keys: CodingKeys enum (per key) or keyDecodingStrategy: .useDefaultKeys (default) · .convertFromSnakeCase — capitalise after each _ , drop inner _ , keep leading / trailing ones: base_uri → baseUri, never baseURI · custom.
Dates: default, .deferredToDate = Date's own Codable: a Double, seconds since 1 Jan 2001 (978 387 200 s after 1970) — Unix time needs .secondsSince1970; also .millisecondsSince1970 · .iso8601 · .formatted · .custom.

URLSession	
start	every task is created suspended → resume()
callback queue	delegate calls + completion handlers on delegateQueue; nil → a serial queue the session creates, not main
async	data(from:) (iOS 15) → (Data, URLResponse)
configs	.default (disk cache, keychain) · .ephemeral (RAM only) · .background(withIdentifier:) (system process; force-quit cancels)
numbers	request timeout 60 s, reset by each arriving byte · resource 7 days · 6 connections per host, HTTP/1.1 only
ownership	delegate held strongly until invalidate... else a leak; .shared: no delegate, no custom config, no background
challenge	urlSession(_:didReceive:completionHandler:) → .useCredential · .performDefaultHandling · .cancelAuthenticationChallenge · .rejectProtectionSpace

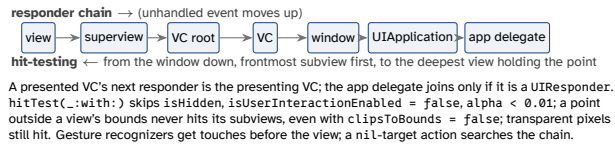
Lifecycle enums & kill codes

UIApplication.State and SwiftUI ScenePhase: .active · .inactive · .background. Scenes opt in via UIApplicationSceneManifest (iOS 13); UIKit logs a message without it from iOS 18.4. TN3187: applicationDidEnterBackground → sceneDidEnterBackground, likewise willResignActive,DidEnterBackground, willEnterForeground. **Hang:** most Apple tools report a main run loop busy > 250 ms. < 100 ms is rarely noticed. 0x8badf00d watchdog · 0xdead10cc file / SQLite lock held while suspending · 0xc00010ff thermal · 0xbba0ca11 VoIP push without a CallKit report.

SOLID — R. C. Martin's own one-liners

S A class should have one, and only one, reason to change. · **O** You should be able to extend a classes behavior, without modifying it. · **L** Derived classes must be substitutable for their base classes. · **I** Make fine grained interfaces that are client specific. · **D** Depend on abstractions, not on concretions.

Touches: down to find, up to handle



Memory & value semantics

	weak	unowned
form	always var, Optional	non-optional or Optional
target dies	ARC sets nil (no observers fire)	access = runtime error; unowned(unsafe) is unchecked
use when	the other lives shorter	the other lives as long or longer

var a = 0, b = 0; let c = { [a] in print(a, b) }; a = 10; b = 10; c() prints **0 10**; list entries are copied when the closure is created; a class reference copies the reference. weak var delegate: P? needs protocol P: AnyObject.

Copy-on-write: arrays, dictionaries, strings share storage until a copy is modified; own structs only by hand — isKnownUniquelyReferenced(...) counts strong refs only, needs synchronisation. static let: lazy, initialised once even across threads; lazy var: no such guarantee. No mutating call on a let struct. deinit: classes, actors and ~Copyable structs / enums (SE-0390, Swift 5.9). Apple: structs by default; classes for Objective-C interop or controlled identity.

Tools: Debug Memory Graph (+ scheme Malloc Stack for backtraces) · Allocations, Mark Generation around one feature · Leaks: unreferenced malloc buffers + root cycles · the simulator never warns or kills for memory.

SwiftUI state

wrapper	iOS	rule
@State	13	owned; also holds an @Observable model
@Binding	13	read / write a source of truth elsewhere (\$x)
@StateObject	13	created once per container lifetime
@ObservedObject	14	an input — never a default value; not for @Observable
@Observable	17	macro; a view updates only for properties body reads
@Bindable	17	bindings into an @Observable

Migration: StateObject → State · EnvironmentObject → Environment + .environment(...) · ObservedObject → plain property · @Published → nothing (@ObservationIgnored opts out). some View: opaque result type, SE-0244, Swift 5.1.

Testing

Pyramid (Cohn, *Succeeding with Agile*, 2009): Unit · Service · UI; inverted = Ice-cream cone. **70/20/10** is Google Testing Blog 2015, "as a good first guess" — not Cohn.

Test doubles (Meszaros, *xUnit Test Patterns*, 2007): **Dummy** passed around, never used — fills parameter lists · **Fake** working shortcut implementation (an in-memory database) · **Stub** canned answers to calls made during the test · **Spy** a stub that also records how it was called · **Mock** pre-programmed expectations = the specification of the calls.

XCTest: class setUp() → per test: setUp() async throws → setUpWithError() → setUp() → test → teardown blocks (LIFO) → tearDown() → tearDownWithError() → tearDown() async throws → class tearDown(); a crash skips teardown. **Swift Testing** (Swift 6, Xcode 16): #expect records and continues; #require throws ExpectationFailedError, try #require(opt) unwraps; parallel by default (XCTest: one by one); traits .disabled .enabled(if:) .bug .tags .timeLimit .serialized; exit tests Swift 6.2, not on iOS.

Patterns & principles — the authors' words

GoF (Gamma, Helm, Johnson, Vlissides, 1994) — 23: **5 creational** Abstract Factory, Builder, Factory Method, Prototype, Singleton · **7 structural** Adapter, Bridge, Composite, Decorator, Facade, Flyweight, Proxy · **11 behavioural** Chain of Responsibility, Command, Interpreter, Iterator, Mediator, Memento, Observer, State, Strategy, Template Method, Visitor.

Adapter Decorator	converts an interface into the one clients expect same Interface, adds responsibilities at run time — vs "an explosion of subclasses to support every combination"
Facade Proxy	one <i>higher-level</i> Interface over a subsystem surrogate that <i>controls access</i> : remote · virtual (create on demand) · protection · smart reference
Strategy State	interchangeable algorithms; the client picks, so it must know them object "will appear to change its class"; transitions in the Context or (more flexible) in the states
Factory M. Abstract F.	subclasses decide which class to instantiate — one product families of related objects, no concrete classes named ("Kit")

Template Method = "Hollywood principle: Don't call us, we'll call you". Mediator: many-to-many → one-to-many · Command → queue, log, **undo** (UndoManager, registerUndo) · Observer: NotificationCenter selector observers need no removal since iOS 9, block observers do; AnyCancellable cancels on deinit · Builder's **Director** · Iterator = IteratorProtocol.next() (nil = end) + Sequence.makeIterator() · Coordinator (Khanlou, 2015) · "Dependency Injection": name settled in Fowler's Jan 2004 article (constructor / setter / interface injection) — a technique; DIP is a principle.

Related: Auto Layout · URLSession & networking · Codable in depth · App & view-controller lifecycle · UIKit touches · ARC · Value vs reference · SwiftUI state & data flow · Testing fundamentals · Test doubles · XCTest & Swift Testing · GoF: creational + structural · GoF: behavioural patterns · SOLID · Crashes & symbolication